

WEEK-1

TASK-1:

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```

root@LAPTOP-RJFD8EBT:~# nano file_copy.c
root@LAPTOP-RJFD8EBT:~# gcc -Wall -Wextra -g file_copy.c -o file_copy
root@LAPTOP-RJFD8EBT:~# echo "This is my OSSP system call experiment." > source.txt
root@LAPTOP-RJFD8EBT:~# cat source.txt
This is my OSSP system call experiment.
root@LAPTOP-RJFD8EBT:~# ./file_copy source.txt destination.txt
File copied successfully from 'source.txt' to 'destination.txt'
root@LAPTOP-RJFD8EBT:~# cat destination.txt
This is my OSSP system call experiment.
root@LAPTOP-RJFD8EBT:~# cmp source.txt destination.txt
root@LAPTOP-RJFD8EBT:~# source_fd = open(argv[1], O_RDONLY);
-bash: syntax error near unexpected token `('
root@LAPTOP-RJFD8EBT:~# strace -e trace=operat_read,write,close ./file_copy source.txt destination.txt
operat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
operat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\220\243\2\0\0\0\0"... , 832) = 832
close(3) = 0
operat(AT_FDCWD, "source.txt", O_RDONLY) = 3
operat(AT_FDCWD, "destination.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
read(3, "This is my OSSP system call expe"... , 1024) = 40
write(4, "This is my OSSP system call expe"... , 40) = 40
read(3, "", 1024) = 0
close(3) = 0
close(4) = 0
write(1, "File copied successfully from 's"... , 64File copied successfully from 'source.txt' to 'destination.txt'
) = 64
+++ exited with 0 +++
root@LAPTOP-RJFD8EBT:~#

```

TASK-2:

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system 3 calls and identify the kernel services involved

[illegible]

```

class(3) = 0
open(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 3
read(3, "1 locale name alias data base\n...", 4096) = 2996
read(3, "...", 4096) = 0
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-MEASUREMENT", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-TELEPHONE", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-ADDRESS", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-ADDRESS", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-NAME", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-NAME", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-PAPER", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-PAPER", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-MESSAGES", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-MESSAGES", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-MSGFORMAT", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-MSGFORMAT", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-COLLATE", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-COLLATE", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-LTIME", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-LTIME", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-NUMERIC", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-NUMERIC", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-CTYPE", O_RDONLY|O_CLOEXEC) = -1 EIOENT (No such file or directory)
open(AT_FDCWD, "/usr/lib/locale/c-UTF-8-CTYPE", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
read(AT_FDCWD, "sample.txt", O_RDONLY) = 3
read(3, "This is a sample file for OSSP s-...", 32768) = 52
write(1, "This is a sample file for OSSP s-...", 51) = 51
read(3, "...", 32768) = 0
close(1) = 0
close(2) = 0
exit_group(0) = 7
root@LAPTOP-RJFDEBWT:~#

```